

Lab03_Teixeira

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Importação dos arquivos

```
# a.
airports <- read_csv(
  "airports.csv",
  col_select = c(IATA_CODE, CITY, STATE, LATITUDE, LONGITUDE),
  show_col_types = FALSE
)

airlines <- read_csv("airlines.csv", show_col_types = FALSE)

processar_chunk <- function(chunk, pos) {
  chunk %>%
    select(DESTINATION_AIRPORT, ARRIVAL_DELAY) %>%
    filter(!str_starts(DESTINATION_AIRPORT, "1")) %>% # c.
    drop_na() %>% # d.
    group_by(DESTINATION_AIRPORT) %>%
    summarise( # e.
      soma_atraso = sum(ARRIVAL_DELAY),
      n_voos = n(),
      .groups = "drop"
    )
}

# b.
estatisticas_acumuladas <- read_csv_chunked(
  file = "flights.csv",
  callback = DataFrameCallback$new(processar_chunk),
  chunk_size = 1000000,
  show_col_types = FALSE
)
```

```
# f.
flights <- estatisticas_acumuladas %>%
  group_by(DESTINATION_AIRPORT) %>%
  summarise(
    media_atraso = sum(soma_atraso) / sum(n_voos),
    total_voos = sum(n_voos),
    .groups = "drop"
  )
```

Operação de “join”

- A coluna que serve de chave na tabela flights é DESTINATION_AIRPORT.
- A coluna que serve de chave na tabela airports é IATA_CODE.

```
# c.
aeroportos_ausentes <- anti_join(
  flights,
  airports,
  by = c("DESTINATION_AIRPORT" = "IATA_CODE")
)
print(aeroportos_ausentes$DESTINATION_AIRPORT)
```

```
character(0)
```

```
# d. & e.
flights <- left_join(
  flights,
  airports,
  by = c("DESTINATION_AIRPORT" = "IATA_CODE")
)
```

Contagem de aeroportos por estado

```
aeroportos_por_estado <- airports %>%
  drop_na(STATE) %>%
  count(STATE, name = "num_aeroportos", sort = TRUE)

print(aeroportos_por_estado, n = 60)
```

```
# A tibble: 54 x 2
  STATE num_aeropostos
  <chr>         <int>
1 TX             24
2 CA             22
3 AK             19
4 FL             17
5 MI             15
6 NY             14
7 CO             10
8 MN              8
9 MT              8
10 NC              8
11 ND              8
12 PA              8
13 WI              8
14 GA              7
15 IL              7
16 LA              7
17 VA              7
18 ID              6
19 WY              6
20 AL              5
21 HI              5
22 IA              5
23 MA              5
24 MO              5
25 MS              5
26 OH              5
27 OR              5
28 TN              5
29 UT              5
30 AR              4
31 AZ              4
32 IN              4
33 KS              4
34 KY              4
35 NM              4
36 SC              4
37 WA              4
38 NE              3
39 NJ              3
40 NV              3
```

41	OK	3
42	PR	3
43	SD	3
44	ME	2
45	VI	2
46	AS	1
47	CT	1
48	DE	1
49	GU	1
50	MD	1
51	NH	1
52	RI	1
53	VT	1
54	WV	1

Mapa de atrasos com leaflet

```
# a. & b. & c.
this <- leaflet(data = drop_na(flights, LATITUDE, LONGITUDE)) %>%
  addTiles() %>%
  addMarkers(
    lng = ~LONGITUDE,
    lat = ~LATITUDE,
    popup = ~paste0(
      "<b>Aeroporto:</b> ", DESTINATION_AIRPORT, "<br>",
      "<b>Cidade:</b> ", CITY, ", ", STATE, "<br>",
      "<b>Atraso Médio:</b> ", round(media_atraso, 2), " min"
    )
  )

# d.
saveWidget(this, file = "mapa.html", selfcontained = TRUE)
this
```

```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-08-31 17:15:57 -03"
```